

ENTRY 037 The First ML Baseline — August 16, 2026 — Status: COMPLETE

Entry 037 — Phase 1 Closes, Phase 2 Opens: Can a Learned Model Beat the Closed Form?

A Tabular Baseline First, Not a GNN -- 238 Circuits Isn't Enough Data for One Yet

1. Why a Tabular Model, Not a GNN, First

The original plan (Entry 033's close) was a GNN over the chip's connectivity graph. 238 labeled circuits is not enough to train a graph neural network without either overfitting or leaning so hard on regularization that any comparison to v4.1 is meaningless -- GNNs typically want thousands of labeled graphs, not hundreds. This entry instead trains gradient-boosted trees and a random forest on the same scalar features v4.1's formulas already use (hop distance, real edge/gate counts from the actual transpiled route, worst edge error and worst T1 along it, plus v4.1's own prediction as an input feature), evaluated with 5-fold cross-validation. If a model can't beat v4.1 with today's dataset, a bigger architecture won't either -- this is the honest floor to clear before scaling up.

2. Result: Roughly Tied Overall, Modest Gains Where It Matters

model	CV MAE (all 238)	CV R ²	MAE on the 12 floor-collapse cases
v4.1 (closed form)	6.29 pts	0.605	20.85 pts
Gradient boosting	6.48 pts	0.594	16.21 pts
Random forest	6.39 pts	0.617	18.81 pts

Across the whole dataset, both learned models land within noise of v4.1's own accuracy -- not a surprise, since v4.1's prediction is itself one of the input features and dominates the model's learned importance (87.3%). On the 12 floor-collapse cases specifically -- exactly where Entry 036 showed the closed form has a structural ceiling -- gradient boosting cuts the error from 20.85 to 16.21 points, about 22% relative. Real, but far from solved: the model has seen roughly 2-3 floor-collapse examples per cross-validation fold, nowhere near enough to learn the pattern Entry 036 couldn't even isolate to a single channel.

3. What the Model Actually Learned

Feature importances confirm the honest read: v4_1_prediction (87.3%) dominates, with n_2q_gates_total, worst_edge_error, worst_t1_us, n_real_edges_used, and bfs_hop_distance splitting most of the remaining 12.7% roughly evenly. The model is best described as learning a small correction on top of v4.1 rather than an independent prediction -- which chip and circuit kind (Bell vs GHZ) contributed almost nothing (under 1% combined), consistent with Entry 035's finding that the floor-collapse pattern isn't tied to chip identity in any way these coarse features capture.

Entry 037 — Conclusion

The first ML baseline does what a first baseline should: it roughly matches the physics model everywhere the physics model already works, and shows a real but modest edge exactly where the physics model is known to structurally fail. That is evidence the learning approach is pointed at the right problem, not yet evidence it has solved it. 238 circuits, 12 of them floor-collapse cases, is too thin a sample for a tree model -- let alone a GNN -- to do much better than this.

4. Next Step

Grow the batch dataset (Entries 034/035's scripts are already resumable and checkpointed) toward the low thousands of circuits, specifically oversampling short routes on Sherbrooke where floor-collapse concentrates, then revisit both the tabular baseline and a proper GNN over the chip connectivity graph with enough floor-collapse examples per fold to actually learn the pattern Entry 036 could isolate but not explain.

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Entry 037	The First ML Baseline (gradient boosting vs. v4.1, 238 circuits)	Aug 16, 2026	Complete

Script: entry037_ml_baseline.py. Data: quantumbridge_data/entry037_ml_baseline.json. Opens Phase 2 (ML) of the project. Ties v4.1 overall, ~22% relative MAE reduction on floor-collapse cases. Next: grow the dataset before scaling to a GNN.